PYTHON PROGRAMMING INTERNSHIP

TASK 3 PROJECT REPORT

Project Title: Employee Management System with SQLite Database

Submitted By: Palak Mishra

Course: B.Sc. Information Technology

College: GH Raisoni College of Arts Commerce And Science Mangalwari Campus

Internship Domain: Python Programming

Technologies Used: Python, SQLite, REST API, JSON

Academic Year: 2026–2027

1. Introduction

The Employee Management System is a Python-based application developed to manage employee records using an SQLite database. It allows users to add, view, update, and delete employee details through a menu-driven interface.The project also integrates a weather REST API to display the current temperature and humidity of Nagpur. JSON is used to process and store the API response. A logging system records important application activities.

2. Objectives

- To create and manage an employee database using SQLite.
- To perform CRUD operations on employee records.
- To integrate a public REST API using Python.
- To process and store API data in JSON format.
- To implement logging for application activities.
- To organize the project into separate Python modules.

3. Project Structure

The project is divided into different folders and files to make the application organized, easy to understand, and maintain.

- **main.py:** Runs the main menu and manages employee operations.
- **database/db.py:** Connects to the SQLite database and creates the employee table.
- **api/weather_api.py:** Fetches Nagpur weather data from a public REST API.
- **api/weather.json:** Stores the weather API response in JSON format.
- **utils/logger.py:** Configures the logging system.
- **employees.db:** Stores employee records.
- **app.log:** Stores application activity logs.
- **requirements.txt:** Contains the external Python package required by the project.
- **README.md:** Provides project information and setup instructions.

4. Technologies Used

Python: Used to develop the application and implement its functionality.

SQLite: Used to store and manage employee records.

Requests Library: Used to send HTTP requests to the weather API.

REST API: Provides current weather information for Nagpur.

JSON: Used to process and save weather data.

Logging Module: Records important application activities and errors.

Virtual Environment: Keeps the project's Python dependencies separate from other projects.

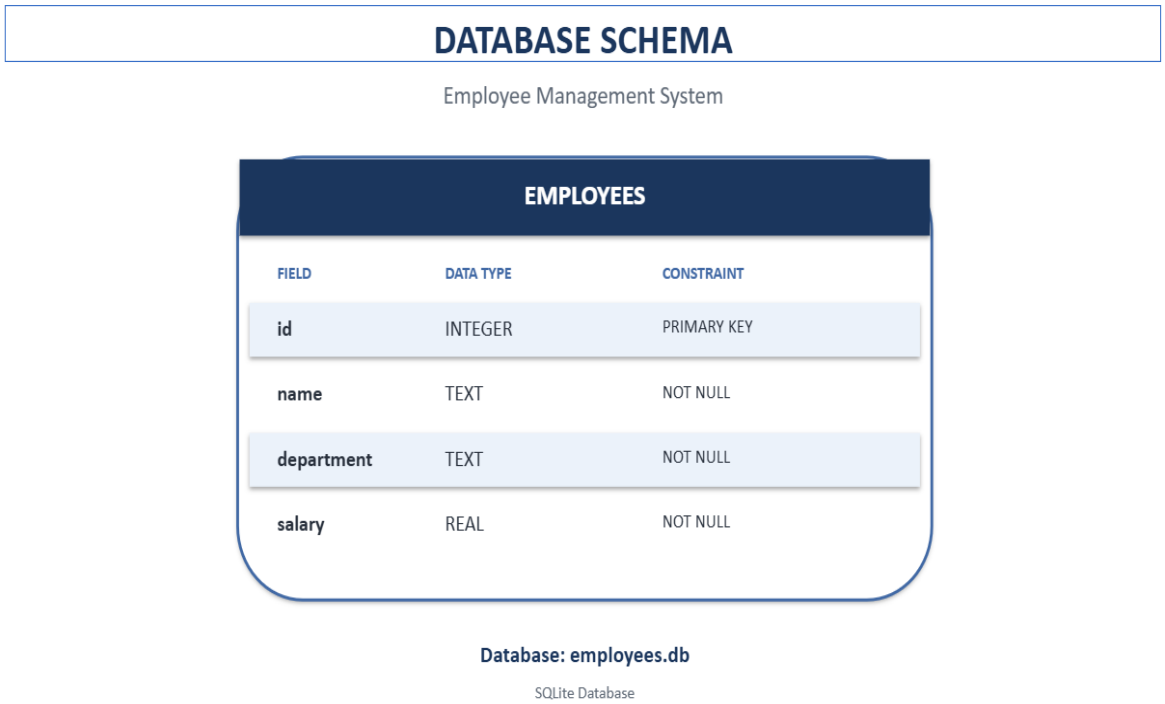


Figure 1: Database Schema of Employee Management System

5. Implementation and CRUD Operations

The Employee Management System uses Python and SQLite to manage employee records. The application provides a menu-driven interface that allows users to perform the following operations.

1. Create (Add Employee): Adds a new employee record containing the employee's name, department, and salary.

2. Read (View Employees): Displays the employee records stored in the database.

3. Update (Update Salary): Updates the salary of an existing employee using their employee ID.

4. Delete (Delete Employee): Removes an employee record from the database using their employee ID.

The application uses SQL queries to perform these operations. Employee records are stored in the `employees.db` database, so the data remains available after the application is closed.

6. Database Description

The project uses an SQLite database named `employees.db`. It contains an `employees` table with the following fields:

- **id:** INTEGER — Primary Key.
- **name:** TEXT — Stores the employee's name.
- **department:** TEXT — Stores the employee's department.
- **salary:** REAL — Stores the employee's salary.

The primary key uniquely identifies each employee record

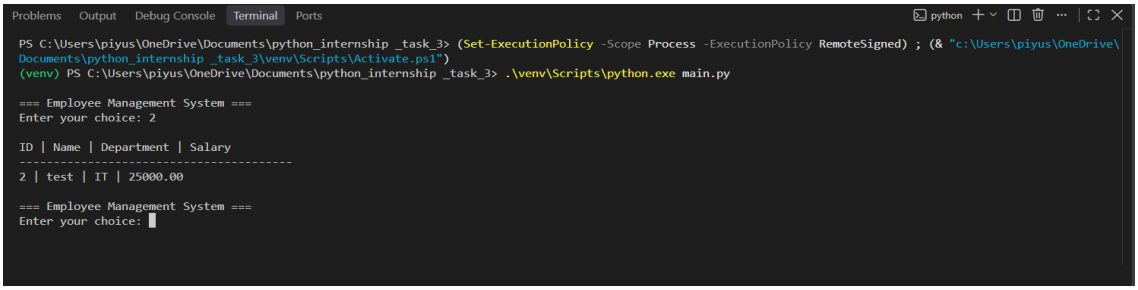


Figure 2: Displaying Employee Records.

7. REST API Integration

The project integrates a public weather REST API to fetch current weather information for Nagpur. The Python `requests` library is used to send an HTTP GET request to the API.The API provides weather information such as temperature and relative humidity. The application processes the response and displays the results in the terminal.If the API request fails, the application handles the error and displays an appropriate message.

8. JSON Processing

JSON (JavaScript Object Notation) is a lightweight format used to store and exchange structured data.In this project, the weather API response is converted into Python data using the `response.json()` method. The weather information is then saved in the `api/weather.json` file using Python's `json` module.The application displays the current temperature in degrees Celsius and relative humidity as a percentage.This feature demonstrates how Python can retrieve, process, and store data received from an external API.

9. Input Validation and Error Handling

The application uses error handling to manage problems that may occur during database operations and API requests. The weather feature handles request errors and invalid or missing response data.These checks help the application respond to errors instead of stopping unexpectedly.



Figure 3: Weather Data Retrieved from the REST API.

10. Application Logging

Logging is used to maintain a record of important events that occur while the application is running. In this project, the Python logging module is used to store application activities in the `app.log` file.The log file records application startup and employee operations such as adding, updating, and deleting employee records. It also helps identify errors and makes debugging easier.

11. Testing

The Employee Management System was tested to verify that its main features work correctly.

Test Case	Expected Result
Add Employee	Employee record is added to the database.
View Employees	Stored employee records are displayed.
Update Salary	Employee salary is updated successfully.
Delete Employee	Selected employee record is deleted.
Check Nagpur Weather	Temperature and humidity are displayed.
Application Logging	Important application activities are recorded in the log file.

12. Conclusion

The Employee Management System was successfully developed using Python and SQLite. It provides basic employee record management through add, view, update, and delete operations. The project also integrates a public weather API to retrieve Nagpur's temperature and humidity.The use of JSON helps process and store weather data, while logging records important application activities. This project helped in understanding database connectivity, modular programming, API integration, file handling, and error handling in Python.

13. Future Scope

The project can be improved by adding employee search functionality, a graphical user interface, login authentication, more employee details, and report generation features.

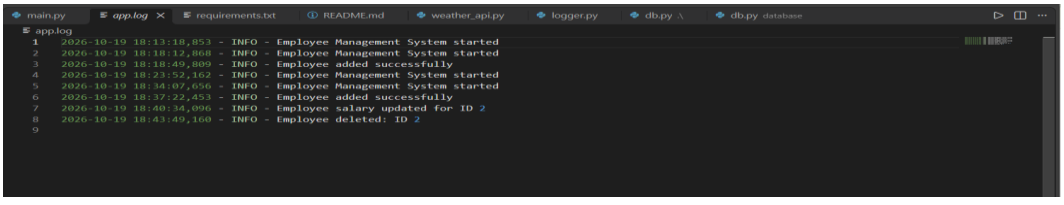


Figure 4: Application Log Recording Important Events.